THE STABLE COMMUTATOR LENGTH OF A RELATOR IS NOT A ONE-RELATOR GROUP INVARIANT

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ABSTRACT. We show that, for the words $r = \mathtt{aabABabABBAbaabABBAb}$ and $r' = \mathtt{aabABabABabABBAbaBAb}$, both in the commutator subgroup of the free group on $\mathtt{a}$ and $\mathtt{b}$, the one-relator groups $\langle \mathtt{a}, \mathtt{b} \mid r = 1\rangle$ and $\langle \mathtt{a}, \mathtt{b} \mid r' = 1\rangle$ are isomorphic, whereas r and r' have distinct stable commutator lengths. This provides a negative answer to a question posed by Heuer and Löh.

Let S be a set and let $F(S)$ be the free group on S. For $w \in F(S)'$, write $\mathrm{cl}_S(w)$ for its commutator length and set

$$\mathrm{scl}_S(w) := \lim_{n\to\infty} \frac{\mathrm{cl}_S(w^n)}{n}.$$

Duncan and Howie proved the gap theorem [3, Theorem 3.3]:

$$w \in F(S)' \setminus \{e\} \quad \Longrightarrow \quad \mathrm{scl}_S(w) \geq \frac{1}{2}.$$

Moreover, stable commutator length in free groups is rational and computable [2]. Heuer and Löh asked whether this quantity can be recovered from the corresponding one-relator group.

Question ([4, Question 1.3(1)])**.** Let S, S' be sets and let $r \in F(S)' \setminus \{e\}$, $r' \in F(S')' \setminus \{e\}$ be relators with $\langle S \mid r\rangle \cong \langle S' \mid r'\rangle$. Does this imply that $\mathrm{scl}_S\, r = \mathrm{scl}_{S'}\, r'$?

For a non-trivial $r \in F(S)'$, put

$$G_r := \langle S \mid r\rangle = F(S)/\langle\langle r\rangle\rangle.$$

Following [4], let $\|G_r\|$ denote the ℓ^1-seminorm of the fundamental class of G_r, called the simplicial volume of G_r. Heuer and Löh proved the universal estimate [4, Corollary 3.12]:

$$\|G_r\| < 4\,\mathrm{scl}_S(r).$$

They also posed the following question.

Question ([4, Question 1.2])**.** For which non-trivial $r \in F(S)'$ does one have

$$\|G_r\| = 4\left(\mathrm{scl}_S(r) - \frac{1}{2}\right)?$$

We give a single example showing that the stable commutator length of a relator is not determined by its one-relator group and, at the same time, that the equality in Question 1.2 need not hold.

Theorem A. *Let $S = S' = \{\mathtt{a}, \mathtt{b}\}$ and*

$$r = \mathtt{aabABabABBAbaabABBAb}, \qquad r' = \mathtt{aabABabABabABBAbaBAb}.$$

The words $r, r' \in F(S)' \setminus \{e\}$ are cyclically reduced and have length 20. Moreover,

$$G_r \cong G_{r'}, \qquad \mathrm{scl}_S(r) = 1, \qquad \text{and} \qquad \mathrm{scl}_S(r') = \frac{1}{2}.$$

In particular, the answer to [4, Question 1.3(1)] is negative.

Corollary 1. *For* $r = \texttt{aabABabABBAbaabABBAb}$,

$$\|G_r\| < 2 = 4\left(\mathrm{scl}_S(r) - \frac{1}{2}\right).$$

Thus the equality in [4, Question 1.2] *does not hold for every non-trivial relator in* $F(S)'$.

We now record a convenient criterion for checking the isomorphism in Theorem A.

Lemma 2. *Let* $r \in F(\texttt{a},\texttt{b})$, $r' \in F(\texttt{x},\texttt{y})$, *let* $p, q \in F(\texttt{a},\texttt{b})$, *and let* $s, t \in F(\texttt{x},\texttt{y})$. *Define homomorphisms*

$$\begin{aligned} \varphi &\colon F(\texttt{x},\texttt{y}) \longrightarrow F(\texttt{a},\texttt{b}), \quad \varphi(\texttt{x}) = p, \quad \varphi(\texttt{y}) = q, \\ \psi &\colon F(\texttt{a},\texttt{b}) \longrightarrow F(\texttt{x},\texttt{y}), \quad \psi(\texttt{a}) = s, \quad \psi(\texttt{b}) = t. \end{aligned}$$

Assume that

(1) $\varphi(r') \in \langle\langle r \rangle\rangle$ *and* $\psi(r) \in \langle\langle r' \rangle\rangle$;

(2) $\psi(p)\texttt{x}^{-1}, \psi(q)\texttt{y}^{-1} \in \langle\langle r' \rangle\rangle$;

(3) $\varphi(s)\texttt{a}^{-1}, \varphi(t)\texttt{b}^{-1} \in \langle\langle r \rangle\rangle$.

Then $\langle \texttt{x}, \texttt{y} \mid r' \rangle \cong \langle \texttt{a}, \texttt{b} \mid r \rangle$.

Proof. The first pair of conditions lets φ and ψ descend to the two quotient groups. The second and third pairs say that the induced maps fix the respective generators after composition, so they are mutually inverse. □

Proof of Theorem A. The two displayed words are visibly cyclically reduced and of length 20. Their exponent sums in both generators vanish, so they lie in $F(S)'$.

Stable commutator length. The identity

$$r' = \texttt{a}\,[\texttt{ab}, \texttt{ABabABabA}]\,\texttt{A}$$

shows that r' is conjugate to a commutator. Hence $\mathrm{scl}_S(r') \le \frac{1}{2}$, and the Duncan–Howie gap gives equality. An exact computation with Walker's `scallop`, implementing Calegari's algorithm, gives $\mathrm{scl}_S(r) = 1$; the reproducible computation is included in [5].

The isomorphism. Relabel r' by $\texttt{a} \mapsto \texttt{x}$ and $\texttt{b} \mapsto \texttt{y}$, so that

$$r' = \texttt{xxyXYxyXYxyXYYXyxYXy}.$$

In Lemma 2, take

$$p = \texttt{BAbaa}, \quad q = \texttt{AABabba}, \quad s = \texttt{YxyXYxy}, \quad t = \texttt{yxYXy}.$$

We verify its three sets of hypotheses. Throughout, $\sim$ denotes cyclic conjugacy, and the initial segment moved in each cyclic shift is shown in blue. Every cyclic conjugate of $r^{\pm 1}$ lies in $\langle\langle r \rangle\rangle$, and likewise for r'; thus each product displayed below is an explicit normal-closure certificate.

First, four of the required certificates are single cyclic conjugates:

$$\begin{aligned} \varphi(s)\texttt{a}^{-1} &= \texttt{ABBAb}\texttt{aabABabABBAbaab} \sim r, &\quad \varphi(t)\texttt{b}^{-1} &= \texttt{AABabbaBAbaBAA}\texttt{BabbaB} \sim r^{-1}, \\ \psi(p)\texttt{x}^{-1} &= \texttt{YxyXYYXyxYXy}\texttt{xxyXYxyX} \sim r', &\quad \psi(q)\texttt{y}^{-1} &= \texttt{YXyxYXX}\texttt{YxyXYxyyxYXyx} \sim (r')^{-1}. \end{aligned}$$

Second, $\varphi(r')$ is a product of three cyclic conjugates of $r^{\pm 1}$:

$$\varphi(r') = \underbrace{\texttt{BAb}\texttt{aabABabABBAbaabAB}}_{\sim r}\ \underbrace{\texttt{abABBAb}\texttt{aabABabABBAba}}_{\sim r}\ \underbrace{\texttt{BAABabbaBAbaBAA}\texttt{Babba}}_{\sim r^{-1}}.$$

Finally, set

$$\begin{aligned} w_1 &= \mathtt{YxyXYxxyXYxyyxYXXYxyXYxyyxYX}, & c_1 &= \mathtt{XYxyXYYXyxYXyxxyXYxy} \sim r', \\ w_2 &= \mathtt{YxyXYxxyXYxy}, & c_2 &= \mathtt{yxYXXYxyXYxyyxYXyxYX} \sim (r')^{-1}, \\ w_3 &= \mathtt{YxyXY}, & c_3 &= \mathtt{xxyXYxyXYxyXYYXyxYXy} = r'. \end{aligned}$$

Substituting the displayed words and cancelling adjacent inverse pairs gives

$$\psi(r) = w_1 c_1 w_1^{-1} \, w_2 c_2 w_2^{-1} \, w_3 c_3 w_3^{-1},$$

and therefore $\psi(r) \in \langle\langle r' \rangle\rangle$. Lemma 2 now yields $G_r \cong G_{r'}$. $\square$

Because stable commutator length is invariant under automorphisms of the free group, conjugation, and inversion, Theorem A also shows that no automorphism of $F(S)$ sends r to a conjugate of $(r')^{\pm 1}$. Thus the isomorphism above only appears after passing to the one-relator quotients.

Proof of Corollary 1. Simplicial volume is an isomorphism invariant, so Theorem A gives $\|G_r\| = \|G_{r'}\|$. By [4, Corollary 3.12],

$$\|G_r\| = \|G_{r'}\| < 4\,\mathrm{scl}_S(r') = 2 = 4\left(\mathrm{scl}_S(r) - \frac{1}{2}\right). \qquad \square$$

Computational Methodology

The search began with cyclically reduced relators in F_2' whose one-relator groups are isomorphic but which do not lie in the same $\mathrm{Aut}(F_2)$-orbit. Claude Desktop with Opus 5 designed and orchestrated an exhaustive search through $\mathrm{Aut}(F_2)$-orbits of words of length at most 20. Candidate pairs were filtered using the first homology of low-index subgroups, Alexander polynomials, and counts of homomorphisms to small finite groups. For candidates not separated by these invariants, we constructed explicit isomorphisms and compared stable commutator lengths with `scallop` [6].

The code, search pipeline, and explicit certificates are available in [5]. The same repository reproduces the computation $\mathrm{scl}_S(r) = 1$ and contains a Lean formalisation of the six identities used above, produced with the assistance of Harmonic's Aristotle [1].

AI use statement. Claude Desktop with Opus 5 assisted in designing and orchestrating the computational search for candidate counterexamples.

Acknowledgement. The author thanks Clara Löh and Nicolaus Heuer for their attention to this solution and for their helpful correspondence. The author is grateful to Vasily Ionin for stimulating discussions and careful reading of a draft of this paper.

References

1. T. Achim et al., *Aristotle: IMO-level automated theorem proving*, 2025, arXiv:2510.01346.
2. Danny Calegari, *Stable commutator length is rational in free groups*, Journal of the American Mathematical Society **22** (2009), no. 4, 941–961, doi:10.1090/S0894-0347-09-00634-1.
3. Andrew J. Duncan and James Howie, *The genus problem for one-relator products of locally indicable groups*, Mathematische Zeitschrift **208** (1991), no. 1, 225–237, doi:10.1007/BF02571522.
4. Nicolaus Heuer and Clara Löh, *Simplicial volume of one-relator groups and stable commutator length*, Algebraic & Geometric Topology **22** (2022), no. 4, 1615–1661, doi:10.2140/agt.2022.22.1615.
5. Artem Semidetnov, *Code, search pipelines and certificates accompanying this paper*, GitHub repository, `https://github.com/s3midetnov/scl-one-relator`.
6. Alden Walker, *scallop*, GitHub repository, `https://github.com/aldenwalker/scallop`.

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